

## 2.

### a. Ideal switches: a description of the important properties of an ideal switch.

- **Unlimited current and voltage capability** - An ideal switch can carry any amount of current when it is ON and can block any amount of voltage when it is OFF. There are no maximum ratings, no breakdown limits, and no thermal restrictions [1], [2].
- **Zero voltage drop when ON and zero leakage when OFF** - In the ON state, the voltage across the switch is exactly 0 V, meaning no power is lost during conduction. In the OFF state, its resistance is infinite, so absolutely no current (leakage) flows [1], [2].
- **Instantaneous switching with no losses** - The switch changes from ON to OFF (and vice versa) instantly, meaning turn-on and turn-off times are zero. Because voltage and current never overlap during switching, there is no switching loss [1], [2].
- **100% efficiency and infinite switching frequency** - Since there are no conduction or switching losses, the switch dissipates zero power and operates with perfect efficiency. With zero switching time, it can operate at infinitely high frequency [1].
- **Negligible control power and perfect reliability** - The switch requires almost no energy to change states, does not generate heat, does not degrade, and can operate indefinitely without failure [1], [2].

### b. Switch non-idealities:

#### i. Qualitatively describe each non-ideality that is expected in a real switch.

- **Limited current rating** - A real switch can only carry a certain maximum current when ON. If the current is too high, the device overheats and may fail [1], [2].
- **Limited voltage rating** - When OFF, the switch can only block a certain maximum voltage. If this limit is exceeded, breakdown or damage can occur

[1], [2].

- **Nonzero ON-state voltage drop** - When the switch is ON, there is a small voltage drop across it. This causes power loss and heating (conduction loss) [1].
- **Leakage current in OFF state** - Even when OFF, a small current may flow due to finite resistance. This causes small power loss [1], [2].
- **Finite switching time** - A real switch does not turn ON or OFF instantly. It takes some time to change states, which limits how fast it can operate [1], [2].
- **Switching losses** - During switching, voltage and current exist at the same time, causing additional energy loss [1].
- **Thermal limits** - Because of energy losses, the switch heats up and must operate within safe temperature limits [1], [3].

## ii. Quantitatively state the behavior caused by each non-ideality

- **ON-state voltage drop (conduction loss)** - Because a real switch has a nonzero voltage drop when ON, power is dissipated as heat. The larger the current or voltage drop, the greater the conduction loss [1].

$$P_{ON} = I_C \cdot V_{CE(sat)}$$

$$I_C = \text{Collector current}$$

$$V_{CE(sat)} = \text{Collector-Emitter saturation voltage}$$

- **Average conduction loss** - If the switch operates periodically, the average power loss depends on how long it remains ON (duty cycle) [1].

$$P_{ON} = I_C \cdot V_{CE(sat)} \cdot d$$

$$d = \text{duty cycle}$$

- **OFF-state leakage current** - A real switch has finite OFF resistance, so a small leakage current flows even when OFF, causing small power dissipation [1], [2].

$$P_{OFF} = I_{LEAK} \cdot V_{SUPPLY}$$

- **Finite switching times** - Since switching is not instantaneous, voltage and current overlap during transitions, leading to energy loss and limiting switching frequency [1], [2].

$$t_{SW(ON)} \neq 0, t_{SW(OFF)} \neq 0, \text{ these are the Turn on and turn off switch time}$$

- **Switching energy loss per cycle** - This represents the energy dissipated during one full switching cycle due to voltage-current overlap [1].

$$W_{SW} = \frac{1}{6} V_{CE(max)} I_{C(max)} (t_{SW(ON)} + t_{SW(OFF)})$$

$$W_{SW} = \text{Energy lost per switching cycle}$$

- **Average switching power loss** - Switching power loss increases with switching frequency  $f$ . Higher frequency means more energy loss per second [1].

$$P_{SW} = W_{SW} \cdot f$$

$$P_{SW} = \text{Average switching power loss}$$

$$f = \text{switching frequency (Hz)}$$

- **Total power loss**- The total power dissipated in a real switch is the sum of conduction and switching losses. This determines heating and efficiency [1].

$$P_T = P_{ON(avg)} + P_{SW}$$

### Sources

[1] M. Alshemmary, *Power Electronics – Fourth Year, Chapter One: General Switching Characteristics*, University of Babylon, pp. 7–12.

[2] O. Ahmed, “Lecture 02 – Features of Ideal and Practical Switch,” *Power Electronics Notes*, 2015.

[3] Ultimate Electronics Book, “Practical Switches,” [Online]. Available: <https://ultimateelectronicsbook.com/switches/>